Padraig Lavery

Software Engineer – Incident Response, Root Cause Analysis & Automation

Belfast, Northern Ireland | LinkedIn | GitHub

PROFILE

Software engineer with 4+ years on the front line of production incidents for high-volume, security-sensitive banking systems in the UK and Middle East, running on AWS and Kubernetes. Triages customer-impacting issues, traces them to root cause, coordinates the fix with other teams, and then automates or documents the response so the next one is shorter or never happens. Building Incident Copilot, an incident-triage tool that suggests likely causes and next steps from runbooks.

EXPERIENCE

Developer | Reward, Belfast | June 2022 – Present

Leading incidents to resolution

- Resolved a live customer-facing outage traced to malformed load-balancer group naming in a Kubernetes/Istio ingress setup, and corrected the Helm deployment templates so the fault could not recur.
- Diagnosed recurring daily out-of-memory crashes in a high-volume transaction ingestion service and removed the cause by loading caches once per batch instead of per file; mitigated further memory-exhaustion failures in ETL jobs.
- Redesigned a month-end invoicing batch with per-partner error isolation, explicit timeouts and safe retry and reconciliation, so a failure on one high-volume partner no longer aborts the whole run.

Making the next incident shorter

- Wrote runbooks and SOPs (re-sending failed outbound files, re-running failed jobs) that let the production services team resolve recurring issues without escalating to engineering.
- Automated scheduled file processing with EventBridge, Lambda and Fargate, removing manual triggers.
- Led a cross-team initiative for automated API testing, adding request validation, shared error handling, OpenAPI docs, unit tests and a stable GitHub Actions CI pipeline.

Security incidents and compliance

- Closed information-leak risks by removing stack traces and SQL details from error responses and standardising input validation across partner microservices.
- Remediated critical Snyk-flagged vulnerabilities for PCI DSS compliance and planned testing for AWS SDK v3 and Node 22 runtime upgrades across repositories and bank deployments.
- Built audit logging of staff access to customer accounts across UK and Middle East bank portals for client security audits.

Team

- Mentored and onboarded four new starters; run the team's retrospectives.

Intern Software Engineer | Flexera, Belfast | May 2020 – April 2021

- Developed a cloud-cost API using Go, Docker, Kubernetes and Git.

PROJECTS

Incident Copilot (in progress) | Python, FastAPI, PostgreSQL + pgvector, Ollama, Docker

- Takes a production incident description, retrieves matching runbook content and suggests likely root causes and next steps, with agentic tool-calling for live service status.

SKILLS

- **Incident management:** triage, root cause analysis, cross-team coordination, runbooks/SOPs, post-fix hardening
- **Cloud and infrastructure:** AWS (Lambda, Fargate, S3, EventBridge), Kubernetes, Helm, Istio, Docker, Terraform, GitHub Actions
- **Languages:** JavaScript (Node.js), C#, Python, Go, SQL
- **Data:** PostgreSQL, MongoDB
- **Security:** Snyk, OWASP, PCI DSS remediation, audit logging

EDUCATION AND CERTIFICATIONS

- BSc (Hons) Computer Science, 2:1 – Ulster University, Belfast, June 2022
- AWS Certified Developer – Associate; AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner – preparing, exam November 2026